

IMPACT STATEMENT FOR ‘OPTIMIZING RAG DOCUMENT RETRIEVAL FOR AGRONOMIC ADVICE’ PROJECT

This project builds a smarter search system for agricultural information. Instead of creating new farming advice, it helps farmers and extension officers quickly find the right document from a collection of trusted agricultural resources. The system compares different retrieval methods, including BM25 and dense embedding search, to determine which approach surfaces the most relevant material for a given question, and measures success using standard metrics such as nDCG@5.

The main benefit is time saved. Extension officers currently have to search manually through long manuals and reports to answer farmer questions. A better retrieval system means faster, more accurate answers, which matters most in rural areas where access to trained agronomists is limited.

The data comes from two respected organizations, IITA and NAERLS, and this shapes what the system knows well. Because NAERLS focuses on Nigeria, the system will generally perform better for Nigerian farming conditions than for other African contexts. The team acknowledges this and refrains from claiming the tool works equally well everywhere.

One notable risk is that since the system returns document excerpts rather than fully explained advice, someone could misread a passage outside its original context, such as applying advice meant for a different season or soil type. To guard against this, the design keeps extension officers involved as the people who interpret and explain results to farmers, rather than letting the system act as a final authority. Every result also keeps its source attached, so users can trace where the information came from.

The dataset is only in English, so the system does not serve non-English speakers directly. Extension officers are expected to translate guidance for farmers, which depends on their continued availability. No new model is trained in this project, so risks around bias in AI-generated text are not a major concern here. The larger risk lies in retrieval mistakes and how carefully results are applied in the field.